2 **(a)** Conditions for equilibrium:

1. The <u>net/resultant force</u> acting on the body <u>must be zero.</u>
   $\sum F = 0$ -- *Translational Equilibrium*

2. The <u>net/resultant torque</u> about any point <u>must be zero.</u>
   $\sum \tau = 0$ -- *Rotational Equilibrium*

No marks awarded if only equations are given.

(b) **(i)**

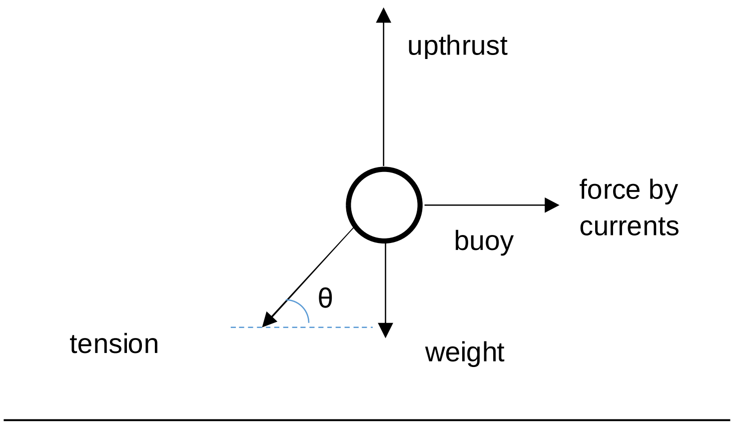

Fig. 2.1

Correct directions (components of the forces should be in all 4 directions) and θ

Correct labelling for the forces identified

Approximate length in both vertical and horizontal directions are the same, both marks awarded. If not, deduct 1.

(ii) $F = kx = 500 \times 2.5 = 1250$ N

(iii) When the buoy was initially at equilibrium, the resultant of the weight and upthrust is equal to the tension by the rope; $T + W = U \rightarrow U - W = T$

$F_{resultant} = kx = 500 \times 1.8 = 900$ N

(iv) The tension by the rope in (b)(ii) is equal in magnitude to the resultant of the force by currents, weight and upthrust

Force by currents = $\sqrt{1250^2 - 900^2}$ = 867.5 N = 868 N (3 s.f.)

(c) $\theta = \tan^{-1}(900/867.5)$

= 46.1° (3 s.f.)

If a different θ is labelled in previous diagram, e.c.f.

3 **(a)** **(i)** Gravitational potential at a point is defined as **work done per unit**